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ADTA-DAST 5340: Discovery and Learning with Big Data

Final Project Report

1. PART I: A Strategy to Employ Machine Learning in a Firm

In this section I assumed that I was hired as a chief information and data officer for Denton Cancer Center that provides diagnosis and treatment services for patients, besides conducting research studies on cancer (Figure 1). So, all assumptions were just performed by me and there is not any external resource for that. However, I was inspired by other cancer centers to form the structure.

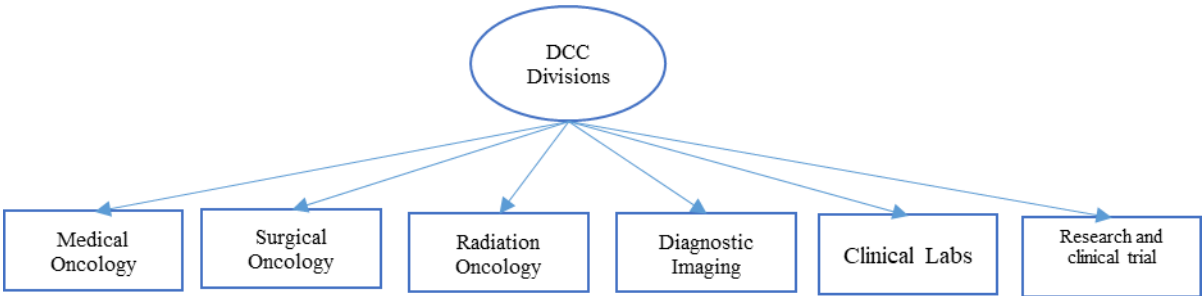


Figure 1 DCC divisions

Machine Learning Deployment Strategy for Denton Cancer Center (DCC)

Item	Details
Domain	Denton Cancer Center
Document Title	Denton Cancer Center AI Strategy
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For	Executive Board of Denton Cancer Center
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1. Executive Summary

Denton Cancer Center (DCC) is a local oncology institution dedicated to the diagnosis, treatment, and research study of cancer. The center employs 150 staff members and is guided by an eight-member

executive board. DCC lives in a competitive healthcare environment that measurable outcomes, operational efficiency, and technological innovation significantly influence institutional reputation and patient trust. Many national and regional cancer centers are rapidly integrating machine learning (ML), and data-driven tools to enhance clinical decision-making, automate the processes, and improve research productivity. In order to being competitive and have a reasonable position among leaders, DCC adopted a comprehensive machine learning strategy and applied technological advancements in the organizational workflows for achieving long term institutional goals. This strategy is designed to improve clinical operations and patients' outcomes using predictive models, decision support systems, AI-powered automation. It also can enhance the results of cancer research studies through using advanced models. This strategy reduces the shortages by optimizing scheduling, allocation of resources and automizing the processes. According to the explanation DCC has enough financial and organizational capacity needed to implement this multiyear strategy plan including modernization of infrastructure, increasing technical staff, and building ML frameworks. For project lifestyle, CRISP DM is used. This document presents the strategy designed for machine learning transformation in DCC

2. Organizational Background and Goals

2.1 DCC Structure

Denton Cancer Center (DCC) has following departments

- **Medical oncology**

This department provides cancer treatments using drugs like chemotherapy, immunotherapy, hormone therapy. Oncologists arrange treatment plans, monitor patients, and perform care.

- **Radiation oncology**

In this section of DCC, they use radiations to kill cancer cells and reduce the tumors.

- **Surgical oncology**

In this department, they perform surgeries to remove the tumors, lymph nodes, or cancerous tissues. They also perform surgical treatments and coordinate with other departments.

- **Imaging services**

They provide diagnostic images for cancer detection including MRI, CT, Ultrasound, etc.

- **Clinical laboratories**

This section performs essential lab tests such as blood and tissue tests or getting samples from patients for other departments.

- **Research and clinical trials**

This section conducts studies to identify new treatments or improve diagnosis.

2.3 Major strategic Focuses

Table 1 Major strategic Focuses

Strategic Goals	Role of Machine Learning Techniques
Improve clinical outcomes	Early detection, prediction of cancer recurrences, provide personalized treatment
Improve Operational capacity	Predictive scheduling, Optimized resource allocation
Improve research outcomes	Identify cancer biomarkers
Reduce the organizational expenses	Process automation, prediction
Increased Patients satisfaction	Enhancing care and outcomes for patients

The major strategic focus of DCC is to improve patient care, enhance the operational processes and increase innovations in the research to find proper competitive position. Machine learning plays important role in reaching these goals. To improve clinical outcomes, machine learning methods can provide early cancer detection, prediction of recurrence risk, and provide recommendations for personalized treatments. To improve the operational capacity, they can perform predictions for time scheduling and optimize the resource allocation that help reduce delays and increase efficiency. Also, it can be very helpful in improving research outcomes by identifying cancer biomarkers and leading to great scientific innovations. From financial perspective machine learning can reduce organizational expenses by automizing the processes and providing accurate predictions. Finally, quality and precision will be improved.

3. Main Objectives

3.1. Mission

The mission of this strategy is to apply machine learning models across all sections of the DCC to deliver optimized care specific to each patient and improve the outcomes.

3.2. Main Objectives

1. Apply machine learning models in all sections of the organization
2. Build an integrated data and analytics platform
3. Create a machine learning operation ecosystem
4. Train the team in machine learning

5. Develop safe, ethical, and responsible AI
6. Increase satisfaction and patients' outcomes

4. DCC Machine Learning Strategy

1. Modernizing our data systems and infrastructure
2. Using the tools and technologies needed for machine learning
3. Enhancing staff skills and building ML expertise
4. Creating clear rules and standards for responsible and ethical AI
5. Implementing ML projects across different departments
6. Using a standardized ML project workflow (CRISP-DM)

The DCC Machine Learning Strategy includes these elements to successfully implement AI across the organization. The first is modernizing the data systems and infrastructure. It is essential for data be stored in secure, interoperable environments. The second component focuses on deploying the right tools and technologies for machine learning, including cloud platforms, MLOps tools, and AI development frameworks to help to build ML models efficiently. So, DCC needs to boost staff skills and develop internal ML expertise by using training programs. An emphasis is on responsible and ethical AI, with clear rules, and documentation. The strategy also includes implementing ML projects across multiple departments. Finally, DCC will apply a standardized ML workflow, such as CRISP-DM, to execute projects from problem definition to data preparation, modeling, evaluation, and deployment, to suggest reliable machine learning solutions.

4.1. Modernizing Data systems and Infrastructure

DCC needs a modern, and secure data foundation to support machine learning across the organization. This strategy shows how data will be collected, stored, transformed, governed, and ultimately used to AI applications.

4.1.1. Data Architecture

The DCC will use a hybrid cloud data system that connects its existing hospital systems with modern cloud platforms like Azure Health Data Services. This setup allows all systems to work together smoothly, keeps patient information secure, and provides enough computing power for analytics and machine learning. The system starts with the main clinical tools such as the Electronic Medical Record, laboratory information system, radiology information system, and medical imaging information system.

All of this data is sent to a large central storage area called an Enterprise Data Lake, where it can be kept safely in its original form. From there, important data is cleaned, organized, and moved into a Data Warehouse (using tools like Databricks or Snowflake) to support reporting, analysis, and machine learning preparation. On top of this, a Machine Learning Platform uses tools like MLflow manage feature creation, model training, and automation. Finally, the Decision Support Layer delivers insights back to clinicians through dashboards, or apps, by giving predictions, alerts, and useful information to support better clinical decisions and improve patient care (Figure 2).

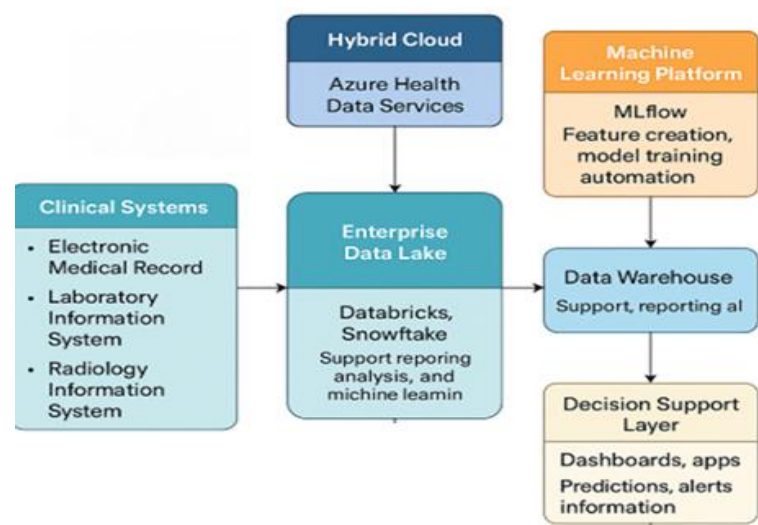


Figure 2 Data Architecture

4.1.2. Data Governance Components

A powerful data governance system can help DCC to keep all clinical and imaging data accurate, private and comply with healthcare rules. A data catalog includes lists of all available datasets so staff can easily find and understand the data they need. For this purpose, DCC should use standard formats like Fast Healthcare Interoperability Resources (FHIR) and Digital Imaging and Communications in Medicine (DICOM) to make sure data from different systems works well together. Besides that, using rules such as Health Insurance Portability and Accountability Act (HIPAA) protect patient information and just allow staffs can view sensitive data (Table 2).

Table 2 Data Governance components

Component	Purpose
Data Catalog	Inventory of all datasets, improving discoverability

Metadata Standards (FHIR, DICOM)	Ensures interoperability across systems
Data Privacy & HIPAA Compliance	Protect patient information
Role-Based Access Control	Restricts sensitive data to authorized users

4.2.Using the tools and technologies needed for machine learning

A powerful pipeline for operating machine learning models helps DCC to build, test, and control machine learning models for clinical use. It can start with version control tools such as Git that help track all the changes to code and models. Then other software or tools can automatically test the models and exclude manual working. DCC will use automated training pipelines that update models when new data arrives to keep the predictions accurate. After developing models, they will be constantly monitored to avoid errors, biases or performance reduction so that the problems can be solved rapidly. All the models will be stored in a model registry to track their versions. These tools can help DCC to have safe, reliable, and accurate ML models for use in the healthcare environment (Table 3).

Table 3 Tools and technologies for ML

Component	Explanation
Version Control (Git)	Tracks all code and model changes
Other Software for ML	Automates testing and deployment of models
Monitoring	Detects errors in live clinical use
Model Registry	Tracks model versions, approval status

4.2.1. ML Technology Ecosystem

The DCC machine learning ecosystem uses modern cloud technology to store data safely, run analytics, and build AI models. Cloud platforms like Azure or AWS provide secure storage and the computing power needed for sensitive clinical data. All types of data such as EHR (Electronic health records), lab results, and medical images are collected in a data lake like Amazon or Azure Data Lake, where they can be stored very cost efficiently. Tools like Databricks or Snowflake are then used to process this data and prepare it for machine learning. Data movement is automated with ETL (Extract, transfer, load) tools such as Airflow. Machine learning models are built using frameworks like TensorFlow, PyTorch, and Scikit-learn. These models are deployed and monitored using MLOps tools such as MLflow. Clinicians receive insights through dashboards created in Power BI or Tableau. NLP tools like

HuggingFace can help analyze clinical notes, while imaging AI tools support tasks like tumor detection and image analysis. Together, these tools create a strong, integrated system that supports advanced analytics and real-world clinical AI (Tabel 4).

Table 4 ML Technology Ecosystem

Layer	Technologies	Purpose
Cloud Platform	Azure / AWS	Secure, scalable computer + storage
Data Lake	Azure Data Lake	Store raw clinical and imaging data
Analytics Engine	Databricks / Snowflake	Large-scale processing and query engine
ELT Tools	Airflow	Automate data flows
ML Frameworks	TensorFlow, PyTorch, Scikit-learn	Build predictive and deep learning models
MLOps Tools	MLflow	Model training, tracking, deployment
Visualization Tools	Power BI / Tableau	Dashboards for clinical & admin insights
NLP Tools	HuggingFace	Extract meaning from clinical text
Imaging AI Tools	Clara Medical	Radiology & pathology AI workloads

4.3.Enhancing staff skills and building ML expertise

A strong clinical machine learning program needs a team of people with different skills working together. The Chief Data Officer or ML Director leads the overall AI strategy. Data Scientists build and test the models, while Machine Learning Engineers deploy the models, so they work smoothly in real hospital systems. MLOps Engineers keep the pipelines running and monitor models for problems. Data Engineers create and maintain the data systems that collect and clean clinical and imaging data. Clinical Informatics Specialists help translate clinical needs into machine learning tasks, so the models are useful to healthcare staff. An AI Ethics ensure all AI work is safe, fair, and follows regulations. Finally, Training Management Leads help staff learn how to use AI tools (Table5).

Table 5 Roles and Responsibilities

Role	Responsibility
Chief Data Officer	Monitor entire AI strategy
Data Scientists	Build predictive models
Machine Learning Engineers	Deploy models into production

MLOps Engineers	monitoring pipelines
Data Engineers	Build data pipelines
Clinical Informatics Specialists	Translate clinical problems into ML solutions
AI-Ethics	Ensure safe, fair, compliant ML practices
Training Management Leads	Manage adoption across staff

A strong training plan is important so everyone in the organization can use and benefit from AI. DCC will focus on AI literacy for clinicians, helping doctors and nurses understand how machine learning works and how to safely use AI tools in care. All staff will be trained on how to handle and protect data to learn good practices for data quality, privacy, and managing patient information. Technical teams, such as data engineers and ML specialists, will have opportunities to earn certifications in Azure, AWS, or Databricks so they can work confidently with cloud and MLOps tools. Leaders will also receive AI-focused leadership training to help them guide change, address ethical issues, and support AI in strategic planning.

4.4. Creating clear rules and standards for responsible and ethical AI

DDC Responsible AI Framework is designed to make sure machine learning is used safely, fairly, and in line with healthcare regulations. The most important goal is patient safety, meaning every model must be carefully tested and monitored before it affects clinical decisions. The framework also supports fairness and transparency, needs clear documentation about how models are built and how they perform. DCC will follow HIPAA rules and stay aligned with new FDA guidelines to make sure AI is used responsibly. To prevent unfair outcomes, the system will regularly check models for bias across different patient groups. Finally, all models must include standardized documentation so their development, validation, and approval can be easily traced and reviewed.

We should meet these objectives:

- Ensure patient safety
- Maintain fairness and transparency
- Comply with HIPAA and new FDA AI/ML rules
- Reduce model bias through audits
- Standardized documentation for all models

4.5. Implementing ML projects across different departments

Machine learning helps many departments at DCC to improve patient care, efficiency, and research. In Clinical Oncology, models can predict treatment response, cancer recurrence, and side effects, help doctors personalize care. In Radiology, AI assists with tumor detection and image segmentation, increasing the speed of diagnosis. In Pathology, AI can analyze digital slides quickly and accurately. Clinical Operations uses ML to predict a patient who misses appointment and improve scheduling and

resource use. In Research, ML helps match patients to clinical trials and identify new biomarkers that support the development of better treatments. Overall, these use cases show how machine learning brings value across the entire organization.

Table 6 ML projects in different departments

Department	ML Use Cases	Value
Clinical Oncology	Treatment response prediction, recurrence risk	Personalized care
Radiology	Tumor detection, segmentation, triage	Faster & safer diagnosis
Pathology	Digital slide classification, morphology analysis	High-speed diagnosis
Clinical Operations	No-show prediction, scheduling optimization	Efficiency
Research	Trial matching, biomarker discovery	Accelerated innovation

4.6.Using a standardized ML project workflow (CRISP-DM)

DCC will follow the CRISP-DM framework to make sure all machine learning projects are safe, reliable, and repeatable. The process starts with Business Understanding, where clinical leaders help identify important problems that AI can solve. Next is Data Understanding, which involves reviewing how complete and accurate the data is. In Data Preparation, the team cleans and transforms the data and creates useful features. The Modeling stage involves selecting algorithms and running experiments. After that, the Evaluation stage checks model performance, fairness, and clinical safety. Finally, in Deployment, the approved model is integrated with the EMR and other clinical systems so it can be used in real care.

4.6.1. Implementation Roadmap

Short-Term (0–6 months)

DCC will use experienced leaders such as the Chief Data Officer and data science leads. The team will build the enterprise data lake, set up the MLOps foundation, deploy the first low-risk ML models, and train the staff.

Mid-Term (6–18 months)

The organization will expand machine learning projects into oncology, deploy imaging AI tools for radiology and pathology, launch NLP tools for analyzing clinical notes, and implement clinical trial–matching algorithms.

Long-Term (18–36 months)

DCC will have a fully operational ML platform, develop advanced oncology prediction models, introduce AI-driven treatment planning, and apply machine learning into everyday clinical and operational workflows.

4.6.2. KPI and Measurement Framework

DCC will measure success using key performance indicators. For clinical outcomes, the goal is to reduce adverse events by 15%. Operationally, the aim is to reduce patient wait times by 20%. Models should achieve accuracy or ROC-AUC above 0.85, and by Year 3, all departments should be using machine learning tools.

2. PART II: Data Pre-Processing

First, I imported essential libraries and modules for data preprocessing including the Panda and NumPy libraries. I also added the seaborn library for creation of heatmap plot. Then I used Panda's data frame to get the file's path and all fields' names. Then, I replaced all the zero values with "nan". By this action, I will treat all as null values and missing data. Especially because zero is not valid and does not have any meaning in those fields. Then, I dropped all the null values by using appropriate method. There were 2, 4, and 5 missing data in fields in "SexUponOutcome", "AgeUponOutcome", and "OutcomeType" respectively. In the next step, I need to make decision on which columns should remain and which of them removed from the dataset. For this purpose, first I used proper methods and loops and convert all categorical variables to the numeric ones. Then using seaborn library, I generated the correlation matrix as it is shown in figure 3.

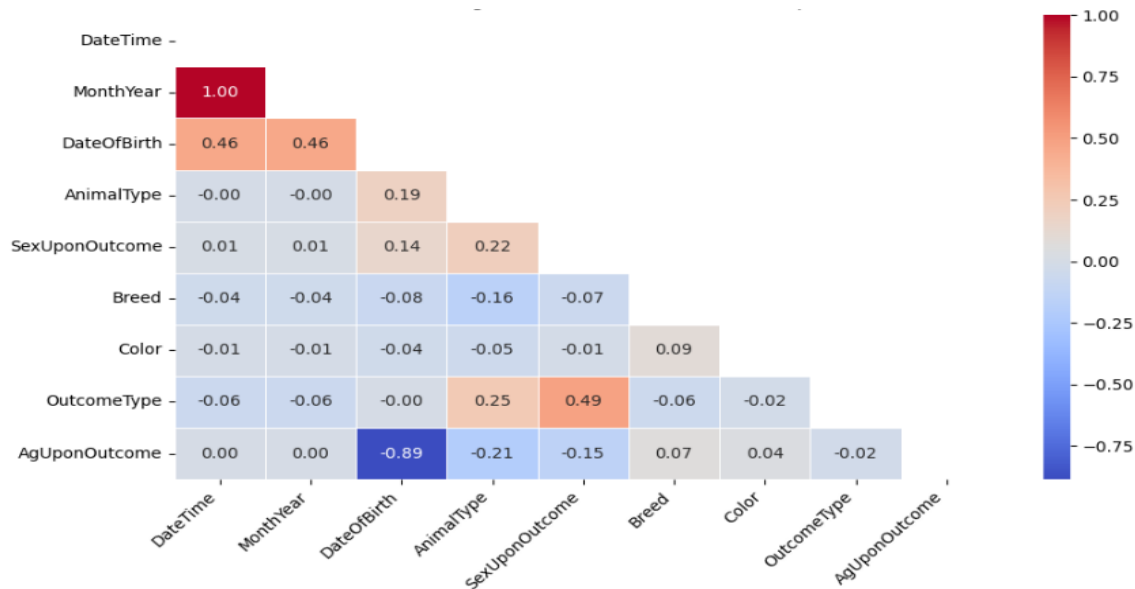


Figure 3 Correlation Matrix of the dataset

As can be observed in Figure 3 in the row named “OutcomeType” all variables except “DateOfBirth” have correlation apposite to 0. Besides that, “DateTime” and “MonthYear” presents similar values for the date of outcome occurrence. We know that always correlation does not indicate causality and it doesn’t make sense to consider date of outcome as one of the predictors for the type of outcome. Therefore, I excluded “AnimalID”, “DateTime”, “MonthYear”, “DateOfBirth” from the dataset and the remaining columns for the clean version of this dataset is “Breed”, “Color”, “OutcomeType”, and “AgeUponOutcome”. Ultimately a Csv named with “Sheltered_animal_clean” extracted.

3. PART III: Build, Train, Test, and Evaluate ML Models

1.1 Data Discovery

The goal of this project is to build a machine-learning model that predicts the age of abalone through using physical attributes such as length, height, diameter, and weight. The dataset contains information about 4,177 abalones and includes 8 predictor attributes plus *Rings*, which is converted to age in years using the formula:

$$\text{Age} = \text{Rings} + 1.5$$

Before modeling, it is important to identify potential data-quality issues, outliers, missing values, and categorical variables. The Sex variable is categorical, and all other features should be numeric. Zero

values in measurements (e.g., length = 0, weight = 0) are biologically invalid and must be treated as missing values.

The whole dataset that we used has 4177 rows and 9 columns, and all columns have numeric values except the sex field that includes three categories: Male, Female, and Infants.

1.2 Data Preparation

First all the columns of the dataset were controlled manually, and the only invalid value was observed in the height field which was recorded as Zero. First these values were replaced by nan to be accounted as missing values and then they were removed from the dataset by “dropna” method. Other values have been controlled and none of them were invalid. So, after dropping null values, others didn’t need any other preprocessing.

After removing the missing values, it contains 4175 abalones that their average length of them is 0.52mm, and they have 0.41mm diameter in average and 0.14 mm height and most of them are small and medium size. All the measurements presented reasonable ranges and the minimum values for the measurements in the dataset are very close to zero. Values related to weight presented significant variations and it ranges from 0.002 to 2.83 g, and shucked weight ranged from 0.001 to 1.49 g. These findings illustrate that this dataset includes abalones of significantly different ages and sizes. Overall, the statistics show that we now have a clean dataset, except some values are very close to zero.

Figure 4, the histograms of all continuous variables in the abalone dataset shows several important features of the data. The length, diameter, and height of the abalones exhibit roughly bell-shaped distributions, and this means they have normal distribution which centered near the medium. However, the height attribute has some values near to zero indicate that some abalones have very small height measurements. Variables relevant to weight include whole weight, shucked weight, Viscera weight, and shell weight displayed right-skewed distributions. This means that most of the abalones are approximately light and few of them are heavy. Often younger abalones are small and light, but mature ones grow large. These visualizations showed that most of the abalones are young. Finally, the Age histogram shows that it is right-skewed and most of the abalones are between 8 and 15 years old and few of them are older than 20 years old. This confirms previous findings.

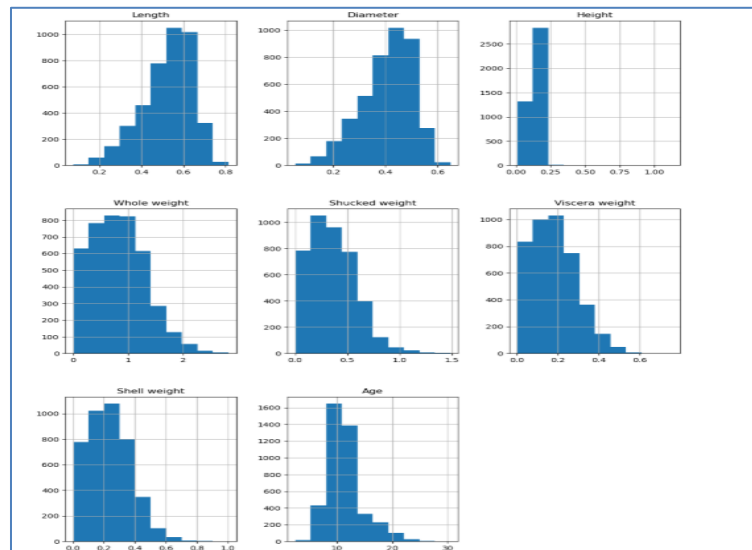


Figure 4 Histogram of numeric fields

Figure 5 shows the kernel density plots of the attributes, and it provides a smooth representation of how each continuous variable is distributed across the dataset. The attributes Length, Diameter, and Height have sharp peaks that show most of the abalones are placed in a narrow range and fewer ones are very small or very large. The density plot of Height shows that their height does not have significant variations. The variables related to weight including whole weight, shucked weight, Viscera weight, and shell weight have right-skewed distributions, and showed that majority of the abalones are relatively light. The age distribution is also right skewed that showed great number of young abalones in the dataset.

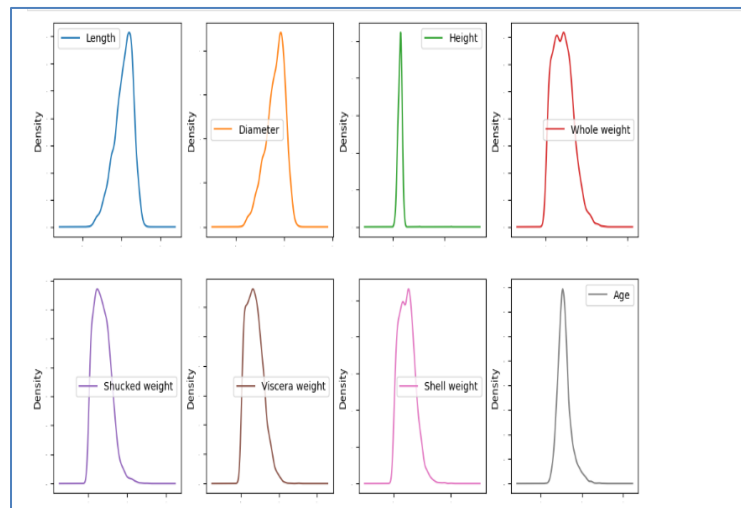


Figure 5 Kernel density plot of abalones' attributes in the dataset

Figure 6 shows that most biological measurements in the dataset are right skewed and have upper outliers. Length and Diameter displays more symmetric distributions and have some lower end outlier. Height is mostly placed around small values and includes unusual large outliers. All weight features have right skewed patterns and most observations were in moderate range. The age of the abalones mostly is between 9 and 12 years with some outliers older than 12.

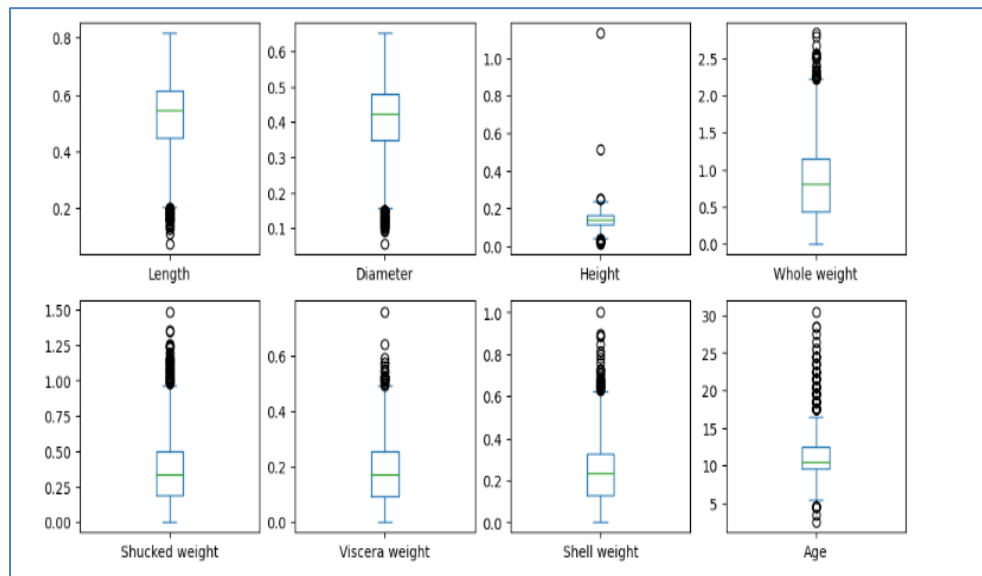


Figure 6 Box plots of the abalones' attribute

Figure 7, the scatterplot matrix shows strong positive relationships among the attributes. All of the variables have strong positive relationships with each other. For interpretation of these relationships, it should be stated that when abalones have higher heights, they will have larger diameters, then they will be heavier and they are older. These relationships are reasonable. Therefore, there are strong positive relationships between height and diameter, between height and weight and between height and age. Also, there is strong relationship between diameter and weight, diameter and age. This relationship is also true about weight and age.

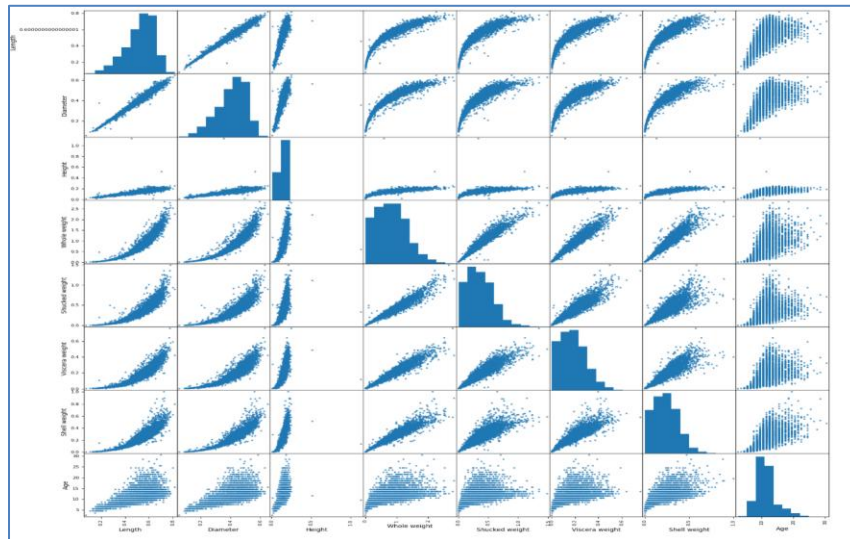


Figure 7 Scatter matrix plot of the attributes of abalones

1.3 . Model Planning

All of the height, diameter, weight, and length of abalones are selected as the predictors of their age. As also it was stated that all of them have strong relationships with age, and the target variable in this project is age. Since the predictors are all numeric, we don't need to transform them into numeric values. Also, since the target variable is numeric, we can use just regression models for prediction.

The models which were used here are linear regression and decision trees, because we want to predict age that is a numeric variable and we need to use regression model. For this problem types of regression models such as Random Forest, or support vector machine can be used as well.

1.4 Model Building

For creating the model, the scikit learn library has been used. First, I split the dataset into the predictors and target variable using the NumPy array and slicing. The train set was built by including 67 percent of the data, and the test set also included 33 percent of the whole data. The selection of the train and test sets became randomized by using random seeds. Also, we calculate the R-square of both models besides generating the intercept and coefficients of the linear regression model.

1.5 Operationalize

The linear regression model and decision tree used to train 67 percent of the data, and the results were extracted on the test set. The coefficients and interceptions of the model were also extracted. For accuracy, I used R-square to present how much of the variability in the target variable is explained by

the predictors. Also, for improving the accuracy of the model and for its evaluation, a cross validation with 10-Fold has been used.

1.6 Communicate the Results

The predictors of the model include sex, diameter, height, weight (whole weight, Viscera weight, shell weight, and shucked weight) that includes 8 first predictors, and age was the final variable and was assumed as target variable.

The R-square of the linear regression model was calculated as 0.49 but the decision tree was 0.14 and much lower than linear regression. Therefore, the linear regression model has better performance and was selected for future predictions and cross validation (Figure 8).

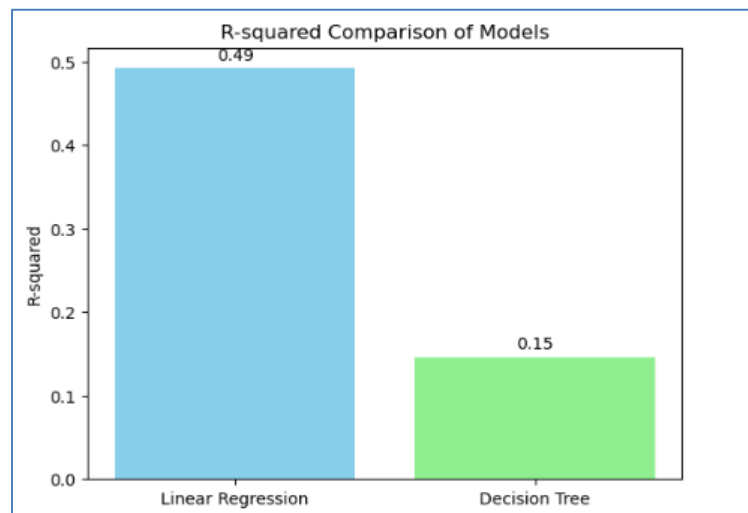


Figure 8 R-Square comparison of two models

According to the results obtained the interception of the model is 4.7 and this means that age of all abalones starts with 4.7 years and according to their own features this amount might become lower or higher. Table1 presents the coefficients of the model for each attribute. The result in this table shows that height and shucked weight have the most influential variables and sex is least influential variable in this project (Table 8).

Table 7 Coefficients of the predictors in linear regression

Variable	Coefficient
Diameter	7.162

Length	1.370
Height	24.251
Viscera weight	-10.439
Whole weight	9.288
Sex	-0.366
Shell weight	6.35
Shucked weight	-20.47

The R-Square of the linear regression model was calculated as 0.49 that means these predictors can only explain about 50 percent of the variability of the age in abalones. This confirms that there are some hidden factors that were not considered in this project that the variance of abalone ages is dependent on them. For examine the results of the model, we consider two records and use the model to predict the age in them. In the first one consider that the sex of the abalone is male and its length is 0.54, and its diameter is 0.4. The height of the abalone was considered 0.15mm, its whole weight was 0.65 gr, its shucked weight 0.2 gr, Viscera weight was assumed 0.15 and its shell weight 0.21. The model predicted its age will be 13.7 years old. This measure will be correct by 50 percent due to the accuracy measured before.

In another assumption consider that the sex of the abalone is female and its length is 0.47, and its diameter is 0.35. The height of the abalone was considered 0.1mm, its whole weight was 0.48 gr, its shucked weight 0.17 gr, Viscera weight was assumed to be 0.081 and its shell weight 0.19. The model could predict that with 50 percent accuracy the age of the abalone is 11.25 years old.

Finally, we the negative mean square error of the cross-validation model with 10-fold for this project was -4.94. Using the following formula the R-square after cross validation calculated to 0.52, and it means that it improved by 0.03.

$$R^2 = 1 - \frac{MSE}{Var(y)} = 1 - \frac{MSE}{3.22^2} = 0.52$$

4. PART IV: Build, Train, Test, and Evaluate ML Models

1.7 Data Discovery

The main goal of this project is to create a machine learning model that can predict the individuals' annual income and whether it is greater than 50,000 or less than or equal to 50000 using demographic attributes such as race, sex, and their marital status or by using other features related to employment and

occupation status information that collected in a census survey. This needs a classification model to identify the level of income for each individual by using those factors. This dataset contains 50000 records and includes attributes such as age, education, occupation, race, sex, their weekly working hours, capital loss and gain besides other predictors. Income was assumed as the target variable. The initial exploring of the variables presented that there are missing and abnormal values in the fields that should be gone through preprocessing. This dataset has both categorical and numeric variables. Age field includes all numerical values and does not contain any abnormal or missing data. Emp_type that shows employment type include different valid types include Federal-gov, Local-gov, Never-worked, Private, Self-emp-inc, Self-emp-not-inc, State-gov, Without-pay However it also includes “?” that should be gone through preprocessing. Fnlwgt related to survey ID and should be removed from the dataset. Education field also includes different levels: 10th, 11th, 12th, 1st-4th, 5th-6th, 7th-8th, 9th, Assoc-acdm, Assoc-voc, Bachelors, Doctorate, HS-grad, Masters, Preschool, Prof-school, Some-college. The marital field includes also these different kinds: Divorced, Married-AF-spouse, Married-civ-spouse, Married-spouse-absent, Never-married, Separated, Widowed. The occupation field presented different categories including: Adm-clerical, Armed-Forces, Craft-repair, Exec-managerial, Farming-fishing, Handlers-cleaners, Machine-op-inspct, Other-service, Priv-house-serv, Prof-specialty, Protective-serv, Sales, Tech-support, Transport-moving. This field also has abnormal “?” data that should be gone through preprocessing. The column titled “Relationship” includes Husband, Not-in-family, Other-relative, Own-child, Unmarried, Wife. Race field also has various categories including: Amer-Indian-Eskimo, Asian-Pac-Islander, Black, Other, White without any observed abnormal values. Sex field also has two types including Male and Female. The field Capital_gain ranges from 0 to 99999 and all are numeric without any invalid data. The field Capital_loss also ranges from 0 to 4356 and all are numeric without need to preprocessing. The field weekly ranges from 1 to 99 and all values are numeric without any invalid value. The field country includes 41 names of different countries, but it also has “?” value that needs to be preprocessed. Finally in the field Income, some records have redundant “.” At their end which need to be removed.

1.8 Pre-Processing Data

In this phase first different groups in categorical variables were explored in order to identify any abnormal values. As explained in previous phase, there are three fields including Emp_type, Occupation, and Country which had “?” values. These invalid measures should be converted to Nan, so that we can treat them as missing values. After this conversion and using “isnull” method, I found 2799 null values in the field employment type, 2809 null values in the Occupation field, and 857 records have

null values in country. All of these null values have been removed from the dataset. The final cleaned dataset has 45222 rows and 14 fields. This dataset includes both numeric and categorical variables. The summary statistics of the dataset show there are 5 numeric variables including Age, Education_num, Capital_gain, Capital_loss, and weekly_hours. On average, the age of the individuals is 38.5 years old with ages ranging from 17 to 90 means that the population includes both young and old people. The average for the highest level of education of the individual is about 10 years of formal education with values ranged from 1 to 16 and shows various educational attainments. The distribution of capital gain and loss are very skewed, and the means are 1101 and 88, and both medians are 0. Weekly working hours presented a pattern of full-time hours with an average of 40.9 hours, median of 40 hours, and a range from 1 to 99 hours. These show both part-time and very long working schedules among respondents.

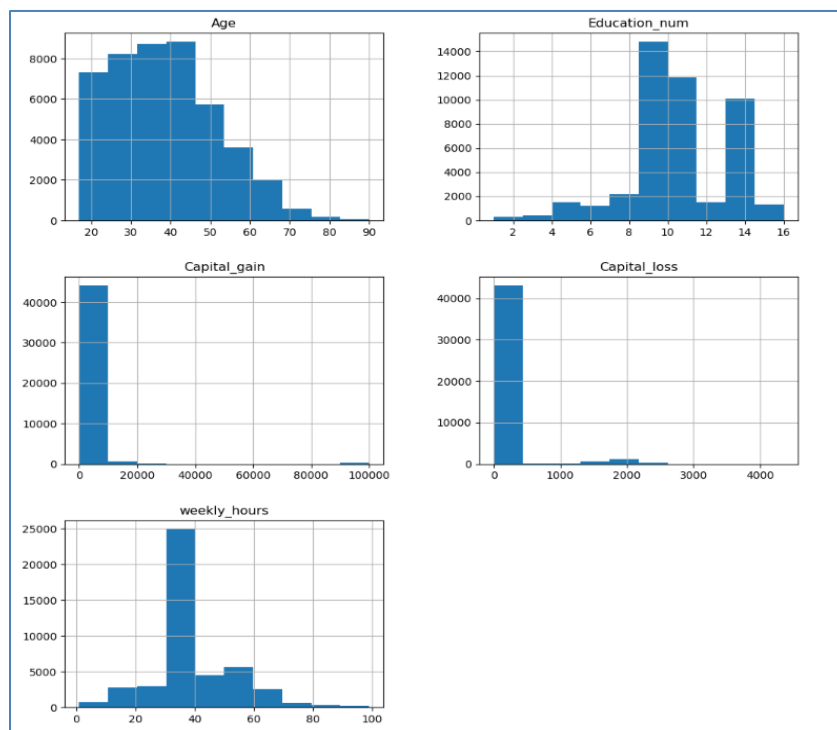


Figure 9 Histogram of Numeric Variables

Figure 9, histograms show the distribution patterns of those five numeric variables in the dataset. Age follows a slightly right-skewed distribution, and most of the individuals are between 25 to 50 years old. The highest education level is more concentrated around 9-13 years of schooling that shows typical high school to early college education levels. Both capital loss and gain are very skewed and large number of observations are clustered close to zero. Weekly hours display a regular pattern which is centered around 35 to 45 hours a week, which shows that most individuals have roughly full-time employment.

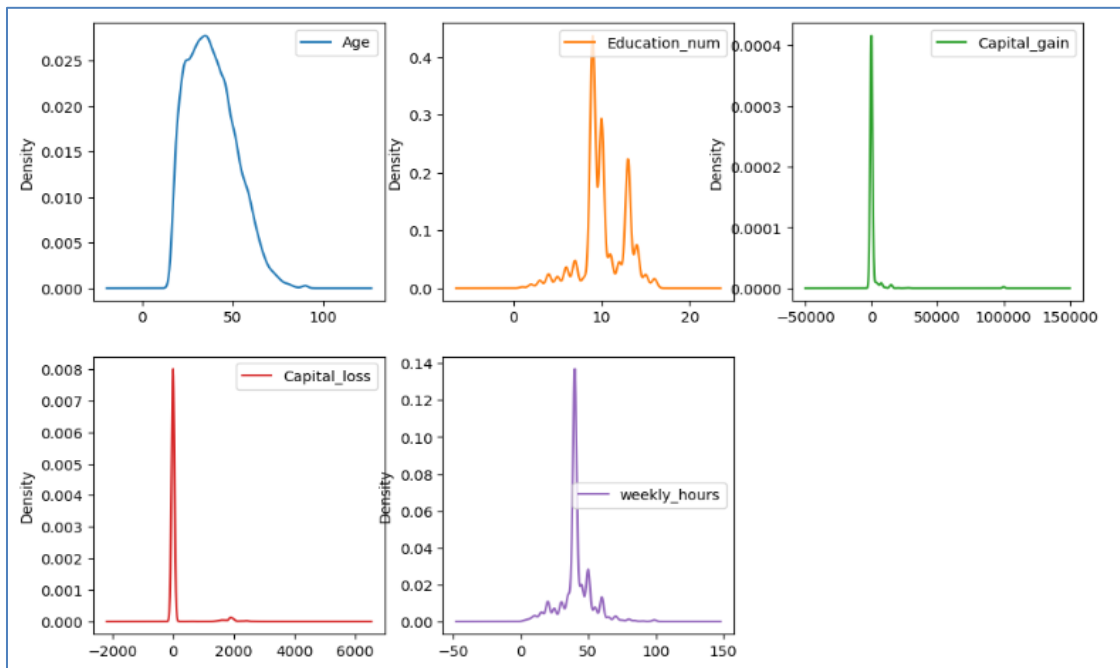


Figure 10 Kernel Density Plot of Numeric Fields

The density plots presented a clearer view on how each numeric variable is distributed in the dataset. Age shows a smooth, unimodal distribution which is concentrated between 25 and 55 years and its peaks are around late 30s to early 40s with very few people who are very young or very old. Education_num shows multiple peaks and reflects that numeric values are assigned to separate levels of education. Most of the individuals have 9 to 13 years of education indicated that the dataset is filled more with high-school and college level individuals. However, Capital loss and gain show skewed distributions concentrated near zero. Finally, the working hours have strong peak at around 40 hours, and few individuals have part-time working hours and some with very long working hours (Figure 10).

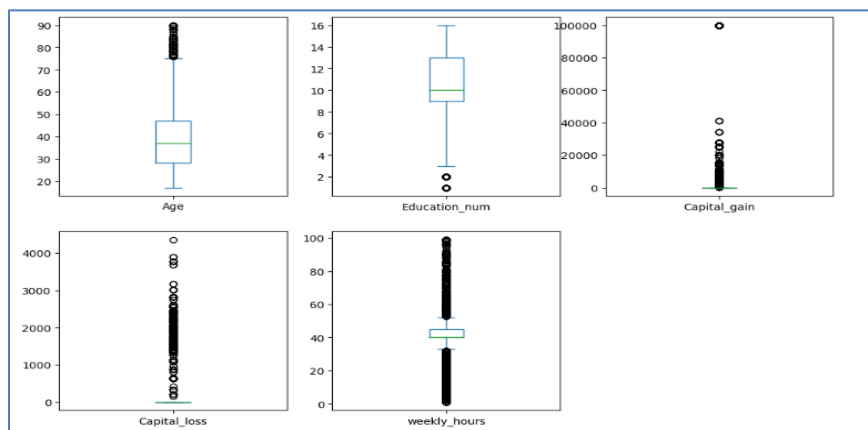


Figure 11 Box plots of numeric fields

Figure 11, boxplots provide important information about the spread and presence of outliers in each numeric variable. Age shows approximately symmetrical distribution centered around mid-30s and only few have high value outliers above age 70. most values in Education_num were placed between 9 and 13 and there were few outliers around the minimal education levels. For capital loss and gain the boxplots are compressed near zero with many outliers in the upper bound. The weekly hours boxplot indicates that most people work between 35 and 45 hours a week and there are numerous outliers for both shorter and very long work weeks reaches up to 100 hours a week.

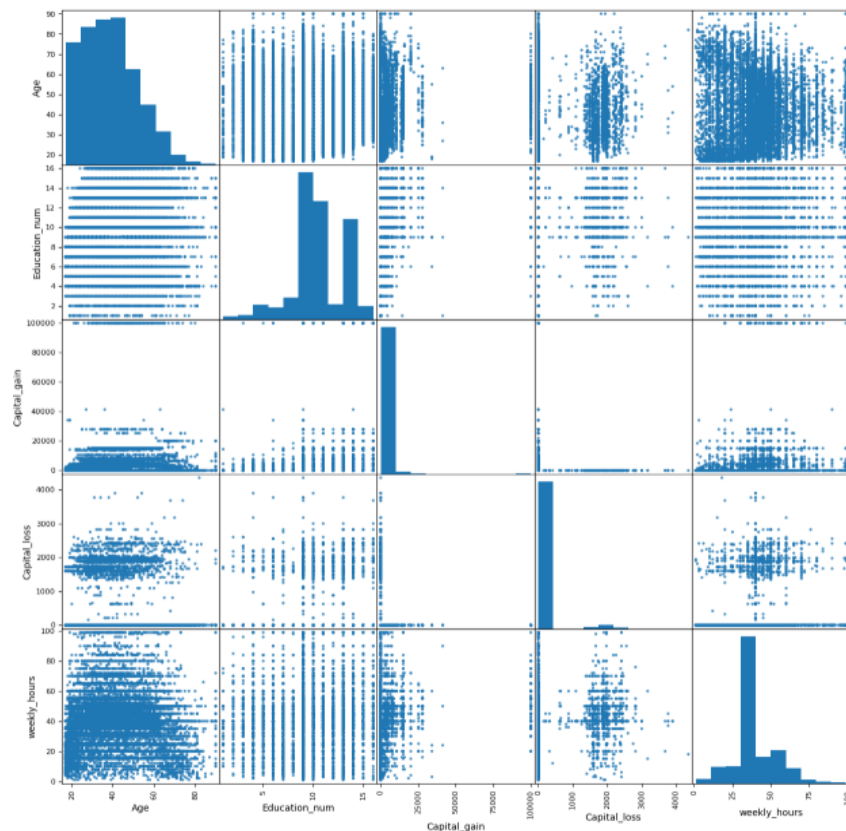


Figure 12 Scatter Matrix plots of the numeric variables

The scatter matrix plot provides complete view of the relationships between the numeric variables including Age, Education_num, Capital_loss, Capital_gain, and weekly_hours. It shows that age does not have any linear relationship with other numeric variables. The highest level of education also shows no strong correlation with capital loss or gain. Weekly hours also do not have any correlation with other numeric variables (Figure 12).

1.9 Model Planning

Since, the main purpose of this project is to build a model to use all current predictors and predict the income level which is a categorical variable. Therefore, I need to use classification models. For this purpose, there are different choices that in this project I use logistic regression and K-nearest neighbors with $k=7$. First, I need to know which models have better accuracy and then select that one for other predictions. Also, I used cross validation with 10 folds to evaluate the model and improve the training process.

1.10 Model Building

For building the model, I split the dataset into the predictors and target values. So, all fields except income were considered as attributes including age, employment type, education level, highest education level, marital status, occupation, relationship, sex, race, capital loss and gain, working hours, and country. Income was also assumed as the response variable. A notable point here is that all categorical variables were transformed into numeric types which made them proper for using in models. Then I created a subset that contains 33 percent test set and 67 percent train set. Both models Logistic Regression and K-nearest neighbors were trained on the train sets and their accuracy were reported on the test set to evaluate which of those models have better performance in classifying records as low income and high-income levels. Finally cross validation with 10 folds were used to both improve the performance of the models and to avoid overfitting.

1.11 Reporting Results

Finally, due to the developed models, the k nearest neighbors' model has better performance in all metrics. KNN model has better precision in classifying individuals with more than 50000 income level. However, it has equal precision in selecting other types of income level. Also Figure 13 presented the comparisons of these two models by comparing their average metrics. As it is illustrated in Figure 13 KNN model could outperform logistic regression in all metrics including precision, recall, F1score, and accuracy. Therefore, this model has been used for prediction of new records.

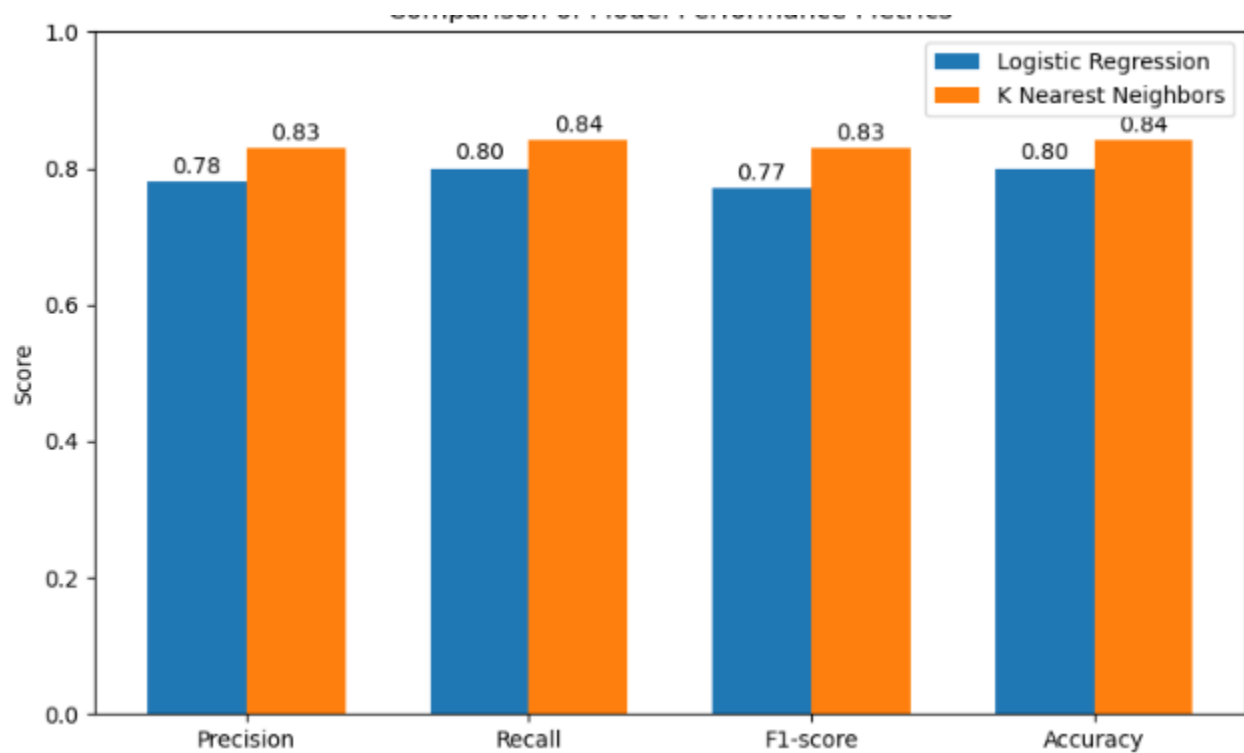


Figure 13 Comparative bar chart for model comparison

So, to have predictions for new records, first I consider two new individuals with their own information. These new records are as follows. The first one is a 42-year-old white male whose employment type is private, and his education level is Bachelor of Science that his highest grade is 13 who is legally married. He is also executive manager and has just relationship as a husband. His capital again is 5178 and his capital loss is 0. He has full time job working about 40 hours a week and he is living in the United States. Both models predicted this individual as a person with income level of more than 50000. Surprisingly in another prediction a black woman whose age is 40 years old, and she has private employment type and has a doctorate degree with highest educational level equal to 16 and is legally married and has just professional occupation that lives in United States and work about 60 hours a week without any capital loss and gain was also classified as an individual with high income level by both models. The logistic regression model could classify it as this class with 80 percent accuracy and K-nearest neighbors found its class with 84 percent accuracy.

Finally, to improve the training of the model and increase the accuracy I used 10-Fold cross validation. The results presented that cross validation can only improve the accuracy of two models by 0.04.

5. PART V: Build, Train, Test, and Evaluate ML Models

1.12 Data Discovery

The main goal of this project is to leverage unsupervised machine learning on the `car_evaluation.csv` dataset to identify natural clusters of car characteristics. The dataset includes categorical variables including the price, cost of maintenance, number of doors, passenger capacity, luggage size, and safety. The objective is to find meaningful groups of cars based on these attributes and then use the developed model to categorize two newly created cars into one of these discovered clusters. This analysis can help to understand market segmentation and patterns of preferences of consumers without using any labeled data. This dataset includes seven attributes, six of those are used as predictors for clustering. These are as follows.

- Price – buying price of the car
- Maintenance – maintenance cost
- Doors – number of doors
- Passengers – passenger capacity
- Luggage – size of luggage boot
- Safety – estimated safety of the car
- Evaluation – target label, not used in clustering

After loading data, we explored its structure, unique values, and data types to understand whether they include any missing data or abnormal values. All attributes are categorical. This means that they should be first encoded to numeric values to make them proper for using K-means clustering.

1.13 Pre-Processing Data

To explore in the dataset and find which variables need preprocessing. Different groups in the categorical variables are extracted. In the field number of Doors, there is a group categorized as “5more” that they were replaced with 5 in the dataset. In the field Passengers also, there is a group called “more” that it is also replaced with value 5. These variables are categorical but after replacement they became numeric and in the exploratory data analysis different numeric plots were created for them. So, these plots are not interpretable and all of them have peak points in the values of each column. It’s because the categorical variables were turned into numeric predictors with fixed points.

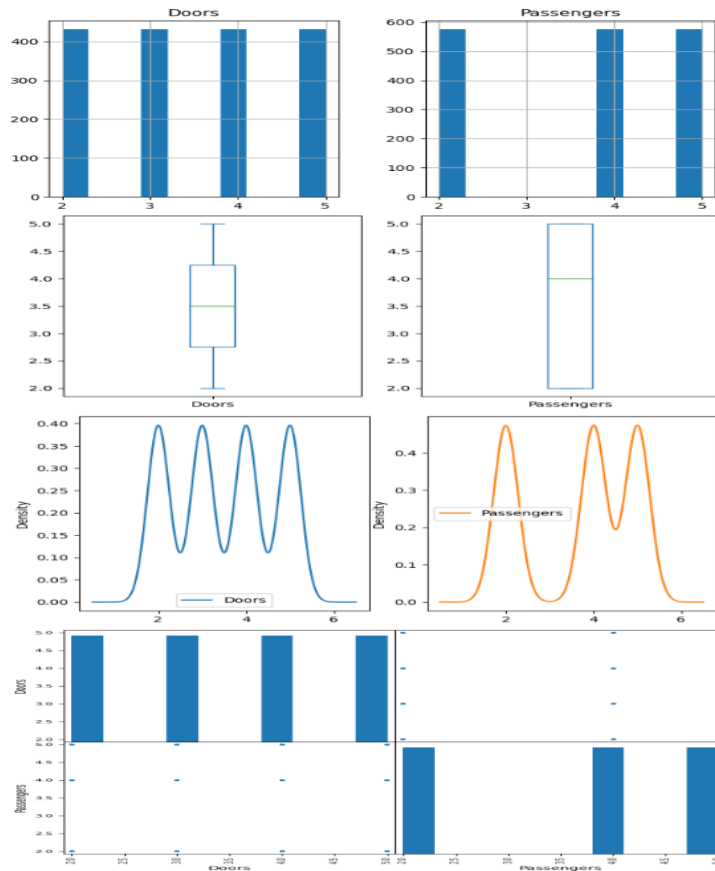


Figure 14 Plots of EDA

These interpretations are also true about scatter matrix plot. As is observable, any relationship between them is not recognizable and interpretable. After that replacement the dataset is ready for further analysis and modeling (Figure 14).

1.14 Model Planning

This project is a clustering problem. For this purpose, I use K-Means clustering for categorizing different cars in the dataset and especially for new predictions. This methodology uses Euclidean distance method for finding the nearest centroid for clustering. All of the variables except evaluation were considered as predictors. Therefore, price, maintenance, doors, passengers, luggage, and safety were taken into account as predictors of clustering. The Evaluation field has been excluded from the matrix of predictors especially because it includes the labels for different clusters. However, I have to decide how many clusters are optimum for this project to create different categories.

1.15 Model Building

Therefore, I used Elbow Methodology for identifying the optimum number of clusters.

In K-Means clustering, the algorithm partitions data into k clusters by minimizing the distances between points and their cluster centroids. Deciding on appropriate k is not an easy task. The Elbow Method helps to find optimal number of clusters by plotting the Within Cluster Sum of Squares (WCSS) regarding the K values and identifying the point where the improvement slows down. This point is called elbow. After finding the optimum K by using this method, I could develop clustering models and use this model to categorize and predict the clusters of new data points. As is shown in Figure 15, the optimal k is 3 especially because after this point, no significant improvements have occurred.

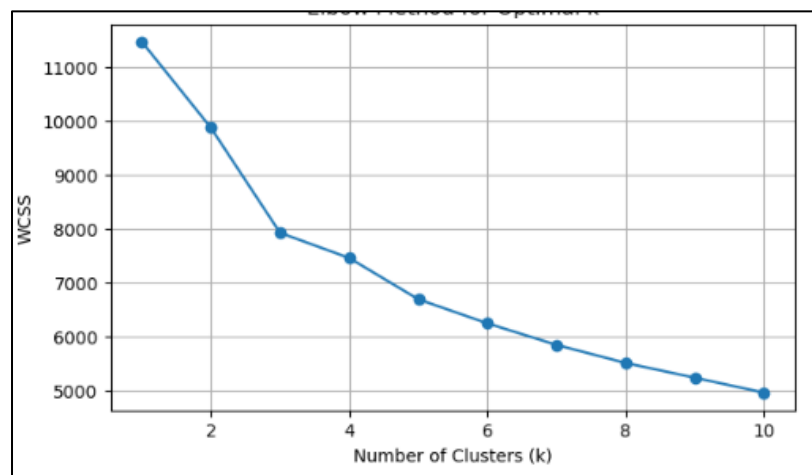


Figure 15 Elbow plot for optimal K

So, using $k=3$, I developed the clustering model to fit on the whole dataset.

1.16 Reporting Results

After development of the model, the centroids for those clusters were printed. These centroids are as follows.

Centroid1: [1.90625, 1.90625, 4.21875, 4.5, 1.0, 1.0]

Centroid2: [1.09375, 1.09375, 2.78125, 4.5, 1.0, 1.0]

Centroid3: [1.5, 1.5, 3.5, 2.0, 1.0, 1.0]

Also, for more clarification the output of cluster assignments produced by the clustering model for the first 10 observations in the dataset were reported. Each number related to the cluster label that the model assigned to a specific data point which clusters identified by integers such as 0, 1, or 2. These labels

show which group each data point belongs to due to their similarity in the features used for clustering. So, I took every 10th element of these labels to find out which clusters they are in. The findings showed that the first data point was placed in the third category, the second data point will be in the first category, and so on.

The scatterplot also cannot display meaningful clusters because it is based on two categorical variables which were encoded to integers. When categorical features such as “Price” and “maintenance” are converted into numeric codes (0,1,2,3) plotting them against each other resulted in only a few reported coordinates and cannot show the clusters well. There is need to apply principal component analysis for having and showing meaningful cluster separation (Figure 16).

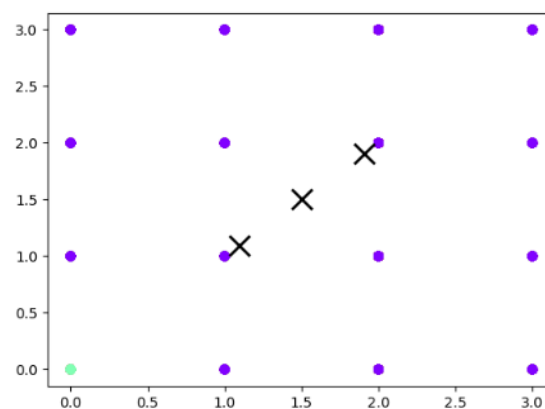


Figure 16 Clustering visualization

Finally, according to the clustering model with $k=3$, two new records have been used for prediction. In the first prediction it was assumed that the car has a medium price, and low maintenance. It has 4 doors and its capacity is 5 passengers. The size of luggage that can be used is small, and there is medium safety for this car. The model finally could decide that it should be categorized as the third cluster. For the second new record, it was supposed that the price of the car is high and it has very high maintenance. This car contains 5 doors, and 5 passengers for the capacity. The luggage that can be placed in this car is big and it has high safety. The model predicted it as the first category

6. PART VI: Evaluate and Compare Machine Learning Models

8.1 Regression Models: Linear Regression vs. Decision Tree (CART) Regression, and Random Forest

For this section, I used the abalone dataset to develop and compare three regression models. The whole dataset first controlled manually, and 2 invalid values were observed including zero values. So, these zero values were replaced by nan values in order to treat them as missing data. Finally, by using

“dropna” method, these values have been removed from the dataset. The processed dataset contains 4175 abalones and 8 predictors including length, height, and weight (whole weight, shucked weight, Viscera weight, and shell weight). The descriptive statistic of this dataset show that their average length is 0.52mm, and they have 0.41mm diameter in average and 0.14 mm height and most of them are small and medium size. All the measurements presented reasonable ranges and the minimum values for the measurements in the dataset are very close to zero. Values related to weight presented significant variations and it ranges from 0.002 to 2.83 g, and shucked weight ranged from 0.001 to 1.49 g. These findings illustrate that this dataset includes abalones of significantly different ages and sizes. Overall, the statistics show that we now have a clean dataset, except some values are very close to zero.

The histograms of all continuous variables in the abalone dataset show several important features of the data. The length, diameter, and height of the abalones exhibit roughly bell-shaped distributions, and this means they have normal distribution which centered near the medium. However, the height attribute has some values near to zero indicate that some abalones have very small height measurements. Variables relevant to weight include whole weight, shucked weight, Viscera weight, and shell weight displayed right-skewed distributions. This means that most of the abalones are approximately light and few of them are heavy. Often younger abalones are small and light, but mature ones grow large. These visualizations showed that most of the abalones are young. Finally, the Age histogram shows that it is right-skewed and most of the abalones are between 8 and 15 years old and few of them are older than 20 years old. This confirms previous findings.

The kernel density plots of the attributes provided a smooth representation of how each continuous variable is distributed across the dataset. The attributes Length, Diameter, and Height have sharp peaks that show most of the abalones are placed in a narrow range and fewer ones are very small or very large. The density plot of Height shows that their height did not have significant variations. The variables related to weight including whole weight, shucked weight, Viscera weight, and shell weight have right-skewed distributions, and showed that majority of the abalones are relatively light. The age distribution is also right skewed that showed great number of young abalones in the dataset.

Boxplot presented that most biological measurements in the dataset are right skewed and have upper outliers. Length and Diameter displays more symmetric distributions and have some lower end outlier. Height is mostly placed around small values and includes unusual large outliers. All weight features have right skewed patterns and most observations were in moderate range. The age of the abalones mostly is between 9 and 12 years with some outliers older than 12.

The scatterplot matrix shows strong positive relationships among the attributes. All variables have strong positive relationships with each other. For interpretation of these relationships, it should be stated

that when abalones have higher heights, they will have larger diameters, then they will be heavier and they are older. These relationships are reasonable. Therefore, there are strong positive relationships between height and diameter, between height and weight and between height and age. Also, there is strong relationship between diameter and weight, as well as diameter and age. This relationship is also true about weight and age.

All of the height, diameter, weight, and length of abalones are selected as the predictors of their age.

Also, since the target variable is numeric, we can use just regression models for prediction.

The models which were used here are Linear Regression, Decision Trees, and Random Forest.

For creating the model, the scikit learn library has been used. First, I split the dataset into the predictors and target variable using the NumPy array and slicing. The train set was built by including 67 percent of the data, and the test set also included 33 percent of the whole data. The selection of the train and test sets became randomized by using random seeds. Also, we calculate the R-square of all models besides generating the intercept and coefficients of the linear regression model.

Also, for improving the accuracy of the models and for their evaluations, a cross validation with 10-Fold has been used.

The R-square of the linear regression model was calculated as 0.49 but the decision tree was 0.17 and for Random Forest 0.55. Therefore, the Random Forest model has better performance than all and Linear Regression has better performance than Decision tree (Figure 17).

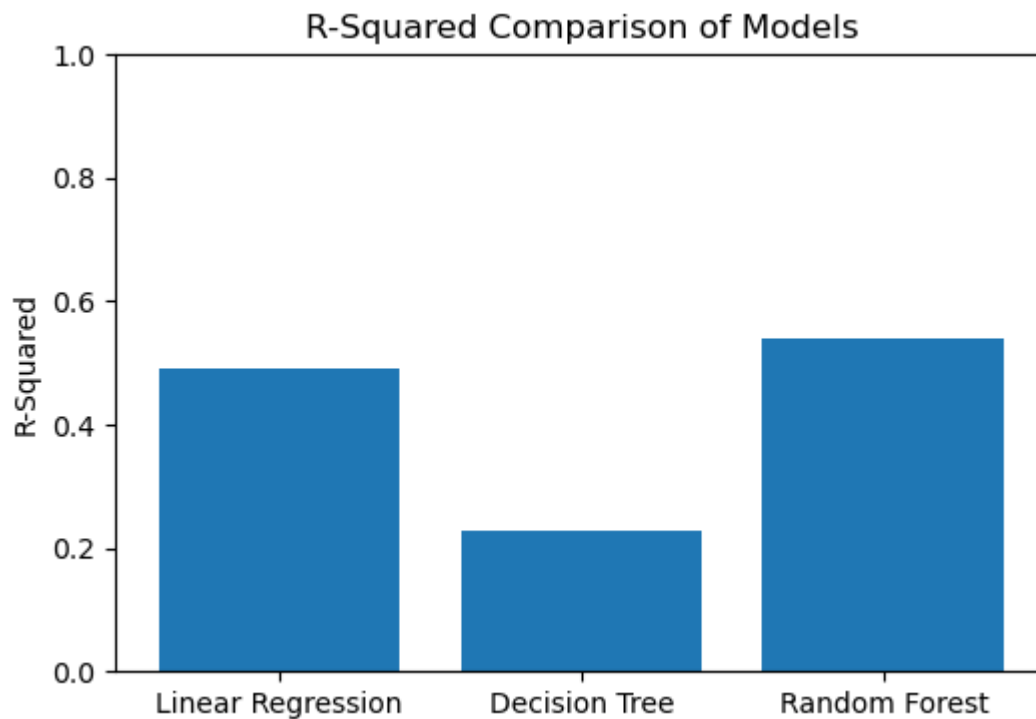


Figure 17 Comparative bar chart of models R-square

According to the results obtained the interception of the linear regression model is 4.7 and this means that age of all abalones starts with 4.7 years and according to their own features this amount might become lower or higher. Table1 presents the coefficients of the model for each attribute. The result in this table shows that height and shucked weight have the most influential variables and sex is least influential variable in this project (Table 9).

Table 8 Coefficients of the predictors in linear regression

Variable	Coefficient
Diameter	7.162
Length	1.370
Height	24.251
Viscera weight	-10.439
Whole weight	9.288
Sex	-0.366
Shell weight	6.35

Shucked weight	-20.47
-----------------------	---------------

The R-square of the Random Forest model is 0.54 that means 54 percent of the whole age variations are described by these predictors. The R-Square of the linear regression model was calculated as 0.49 that means these predictors can only explain about 49 percent of the variability of the age in abalones.

Finally, 17 percent of the whole variation of the age is described by these predictors in decision tree model. This confirms that there are some hidden factors that were not considered in this project that the variance of abalone ages is dependent on them. For examine the results of the model, we consider two records and use the model to predict the age in them. In the first one consider that the sex of the abalone is male and its length is 0.84, and its diameter is 0.4. The height of the abalone was considered 0.16mm, its whole weight was 0.65 gr, its shucked weight 0.2 gr, Viscera weight was assumed 0.16 and its shell weight 0.22. The linear regression model predicted its age will be 14.3 years old. This measure will be correct by 49 percent due to the accuracy measured before. The Decision tree model predicted its age is equal to 10.5 which can correct by 22 percent. Finally, the Random forest model predicted that specific abalone is 13.77 years old which can be accurate by 53 percent. If we want to offer a reliable answer random forest offers the best answer (13.77 years old). If we want to select between Linear regression and Decision tree, we should select Linear regression with more accuracy (14.3 years old)

In another assumption consider that the sex of the abalone is female and its length is 0.48, and its diameter is 0.44. The height of the abalone was considered 0.22mm, its whole weight was 0.94 gr, its shucked weight is 0.2 gr, Viscera weight was assumed to be 0.088 and its shell weight is 0.24. The Random Forest model could predict that with 54 percent accuracy the abalone is 16.8 years old, The Linear regression model predicts with 49 percent accuracy that the abalone is 18.7 years old, and the Decision tree model predicts with 14 percent accuracy that the abalone is 12.5 years old. If we want to offer a reliable answer random forest offers the best answer with higher accuracy (16.8 years old). If we want to select between Linear regression and Decision tree, we should select Linear regression with more accuracy (18.7 years old).

Finally, I used 10-fold cross validation to improve the accuracy of the models and evaluate them. The negative mean square error has been used for evaluation of the models. Using the following formula the R-square after cross validation calculated to 0.53 for Random Forest model (0.01 improvement), 0.52 for Linear regression model (0.03 improvement), and 0.10 for Decision tree model (without any improvement).

$$R^2_{RF} = 1 - \frac{MSE}{Var(y)} = 1 - \frac{4.69}{3.22^2} = 0.547$$

$$R^2_{LR} = 1 - \frac{MSE}{Var(y)} = 1 - \frac{4.93}{3.22^2} = 0.52$$

$$R^2_{DT} = 1 - \frac{MSE}{Var(y)} = 1 - \frac{9.08}{3.22^2} = 0.10$$

Figure 18 presented that after cross validation, random forests outperform other models with minimum mean square error and maximum R-squared.

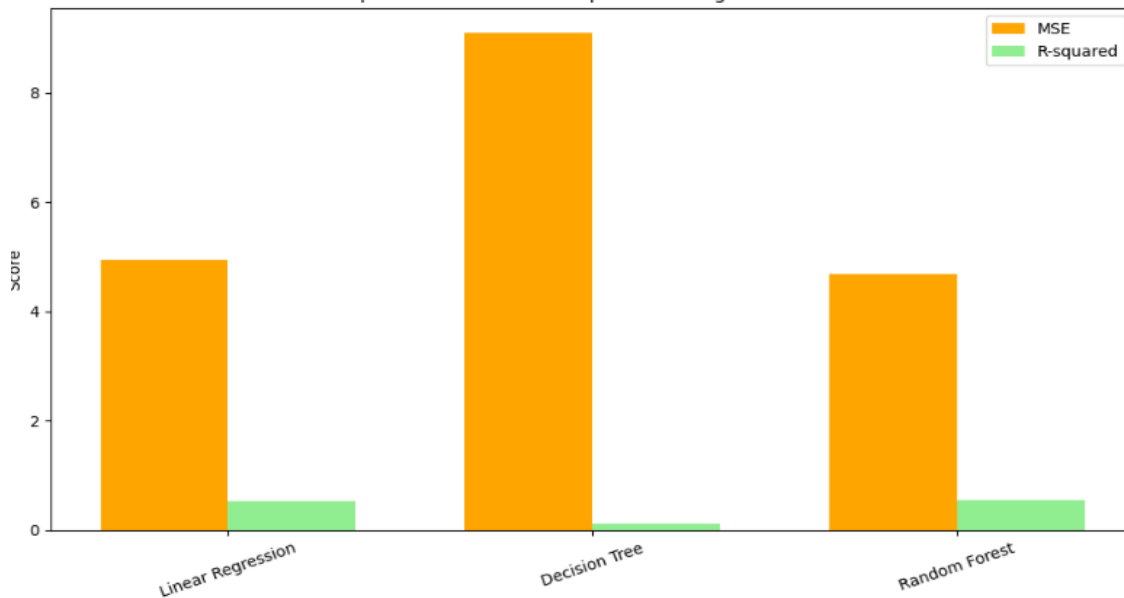


Figure 18 Comparison of MSE and R-squared model for regression models

7. PART VII: Classification Models: Logistic Regression vs. K-Nearest Neighbors vs Random Forest

In this section, the adult_salary dataset containing approximately 50,000 records was used to build a classification model for predicting income level. The dataset includes attributes such as age, education, occupation, race, sex, weekly working hours, capital loss, and capital gain, which were treated as predictor variables, while income served as the target variable. The initial data exploration revealed the presence of missing and abnormal values, indicating the need for preprocessing. The dataset also contains a mix of categorical and numerical variables. The Age attribute is entirely numeric and contains no abnormal or missing values. The Emp_type field includes valid employment categories but also

contains the “?” symbol, which must be handled during preprocessing. The Fnlwgt attribute represents survey sampling weights and was therefore removed because it is not useful for prediction. The Education and Marital fields contain valid categories, while the Occupation field includes both valid categories and the abnormal “?” value, which must be cleaned. The Relationship variable consists only of valid categories. The Sex variable includes the two valid categories Male and Female. Capital_gain (ranging from 0 to 99,999) and Capital_loss (0 to 4,356) both consist of valid numeric values and require no cleaning. Weekly_hours ranges from 1 to 99 and does not contain invalid entries. The Country field consists of 41 country names but also contains the “?” entry, which must be treated as missing. The Income variable also includes some records that display a redundant period (“.”) at the end, which must be removed during preprocessing.

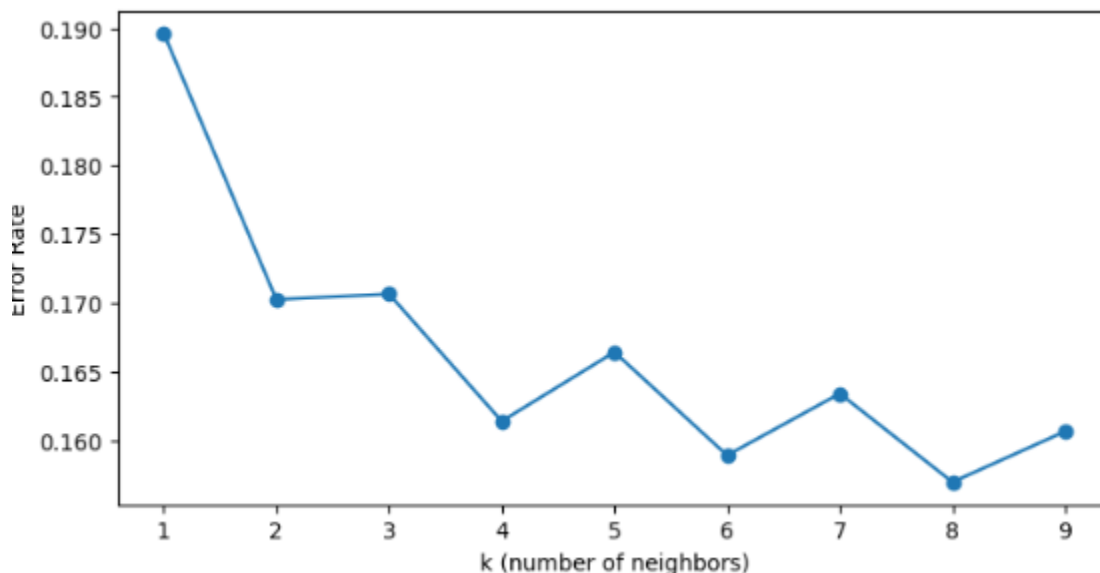
After converting abnormal values to NaN, the isnull() function identified 2,799 missing entries in the employment type field, 2,809 in the occupation field, and 857 in the country field. All records containing these missing values were removed, resulting in a cleaned dataset with 45,222 rows and 14 attributes. The dataset includes both numeric and categorical variables. Summary statistics show that there are five numeric fields: Age, Education_num, Capital_gain, Capital_loss, and Weekly_hours. The average age of individuals is approximately 38.5 years, with values ranging from 17 to 90, indicating representation of both young and older adult populations. The average Education_num is around 10, with values ranging from 1 to 16, reflecting a variety of educational backgrounds. Capital gain and capital loss are highly skewed, with mean values of 1101 and 88, and medians of 0, showing that most individuals did not report gains or losses. Weekly working hours show a typical full-time pattern with a mean of 40.9 hours, median of 40, and a range from 1 to 99 hours, representing both part-time and unusually long work schedules.

The histograms illustrate the distribution of the numeric variables. Age is slightly right-skewed, and most individuals fall between ages 25 and 50. Education_num is concentrated around 9–13 years, indicating common completion of high school or early college levels. Capital loss and capital gain are heavily skewed with most values near zero. Weekly_hours has a strong central pattern between 35 and 45 hours, indicating that most individuals work full-time.

The density plots provide a smoother view of these distributions. Age displays a unimodal distribution peaking in the late 30s to early 40s, with very few extremely young or old individuals. Education_num shows multiple peaks reflecting distinct educational categories. Capital gain and loss remain strongly

skewed toward zero. Weekly_hours has its highest peak around 40, showing typical full-time employment.

The boxplots further illustrate the spread and presence of outliers. Age is relatively symmetrical around the mid-30s, with some outliers above age 70. Education_num mainly falls between 9 and 13, with a few low-level outliers. Capital gain and loss are tightly compressed near zero with many high-value outliers. Weekly_hours shows that most individuals work between 35 and 45 hours, but numerous outliers represent people working both very short and extremely long hours—up to nearly 100 hours per week. The scatter matrix shows the relationships among the numeric variables: Age, Education_num, Capital_loss, Capital_gain, and Weekly_hours. The plots demonstrate that age has no strong linear relationship with education, capital gain, capital loss, or working hours. Education_num similarly shows no meaningful correlation with gain or loss. Weekly_hours also does not correlate with any other numeric variable. This indicates that the numeric predictors are largely independent from one another. To build a classification model that predicts income level, three machine learning algorithms were used: K-Nearest Neighbors (KNN), Logistic Regression, and Random Forest. For the KNN model, the optimal value of k was determined using the Elbow Method. As shown in Figure 16, the error rate and accuracy stabilize around $k = 4$, meaning that increasing the number of neighbors beyond 4 does not improve the model's performance. Therefore, $k = 4$ was selected as the optimal value for KNN.



For building and training the models I split the dataset into the predictors and target values. So, all fields except income were considered as attributes including age, employment type, education level, highest education level, marital status, occupation, relationship, sex, race, capital loss and gain, working hours, and country. Income was also assumed as the response variable. A notable point here is that all

categorical variables were transformed into numeric types which made them proper for use in models. Then I created a subset that contains 33 percent test set and 67 percent train set. All models were trained on the train set and their accuracy was reported on the test set. To evaluate each of those models with better performance in classifying records as low income and high-income levels. Finally cross validation with 10 folds were used to both improve the performance of the models and to avoid overfitting. Finally, due to the developed models,

Table 9 Comparisons between the models

Metric	Logistic Regression	K-Nearest Neighbors	Random Forest
Precision ($\leq 50K$)	0.81	0.86	0.88
Recall ($\leq 50K$)	0.95	0.94	0.92
F1-Score ($\leq 50K$)	0.87	0.9	0.9
Precision ($> 50K$)	0.69	0.75	0.71
Recall ($> 50K$)	0.35	0.53	0.62
F1-Score ($> 50K$)	0.47	0.62	0.66
Accuracy	0.8	0.84	0.84
Macro Avg (F1)	0.67	0.76	0.78
Weighted Avg (F1)	0.77	0.83	0.84

The performance evaluation of these three models presented important differences in how each of these models predicts the income level. Logistic regression had very good performance in predicting individuals with income $\leq 50K$ with precision 0.81 and high recall=0.95 and F1-score=0.87. However, its performance decreased significantly for individuals with income $> 50K$ with precision 0.69, recall equal to 0.35 and F1 score equal to 0.47. KNN could solve this limitation by providing better results for both income levels for the $> 50K$ class, it could achieve a precision=0.75, recall=0.53, and F1-score=0.62 that are higher than logistic regression. It also has strong performance for the $\leq 50K$ class with precision=0.86, recall=0.94, and F1score=0.9. So, the overall accuracy increased to 84 percent, and macro and weighted averages also indicated meaningful improvement. Therefore, KNN can better identify the patterns associated with both low- and high-income individuals and is a more reliable model. Among the three models, Random Forest could reach strongest and the most stable performance. It achieves the highest recall (0.62) and F1-score (0.66) for the $> 50K$ category indicates the best ability to identify high income individuals. It also has very good performance with precision=0.88, recall=0.92, and F1score=0.9 for the $\leq 50K$ class and is equal with KNN in overall accuracy at 0.84. Its macro F1-

score of 0.78 and weighted F1-score of 0.84 are the highest among all models, presenting its strong predictive capability (Table 9).

Then I used two records for predicting with all models. For the first prediction example, the three models produced different outcomes: Logistic Regression and Random Forest predicted that the individual earns more than 50K, while K-Nearest Neighbors predicted less than or equal to 50K. This individual has managerial occupation, college-level education, full-time work at 50 hours per week, and a capital gain of 6,000 which is associated with higher income levels in the dataset. When comparing model performance, Random Forest showed the highest accuracy, macro F1-score, and weighted F1-score among the three models, and it achieved the best ability to correctly identify >50K individuals. Logistic Regression, even if it is not a strong predictor, predicted >50K and provided additional support for this outcome. KNN often predicts lower income for people who may actually earn more because it depends heavily on nearby data points. This makes it less accurate at identifying >50K earners, which is why its recall and F1-scores for that group are lower. Therefore, based on both the individual's feature profile and the superior performance of the Random Forest model, the most reliable prediction for this case is that the individual belongs to the >50K income category.

For the second prediction example, all three models—Logistic Regression, KNN, and Random Forest—agree on the same outcome, predicted that the individual earns more than 50K. This individual is 40 years old, holds a Doctorate degree, works in a professional specialty occupation, is married, works 60 hours per week, and reports a capital gain of 2,000. These characteristics highly align with the patterns associated with higher-income earners in the dataset. Although all models return the same prediction, the Random Forest model remains the most reliable because it showed the highest accuracy, the strongest macro and weighted F1-scores, and the best performance specifically for the >50K class during model evaluation. Therefore, the most reliable prediction for this individual is that he belongs to the >50K income category (Figure 19).

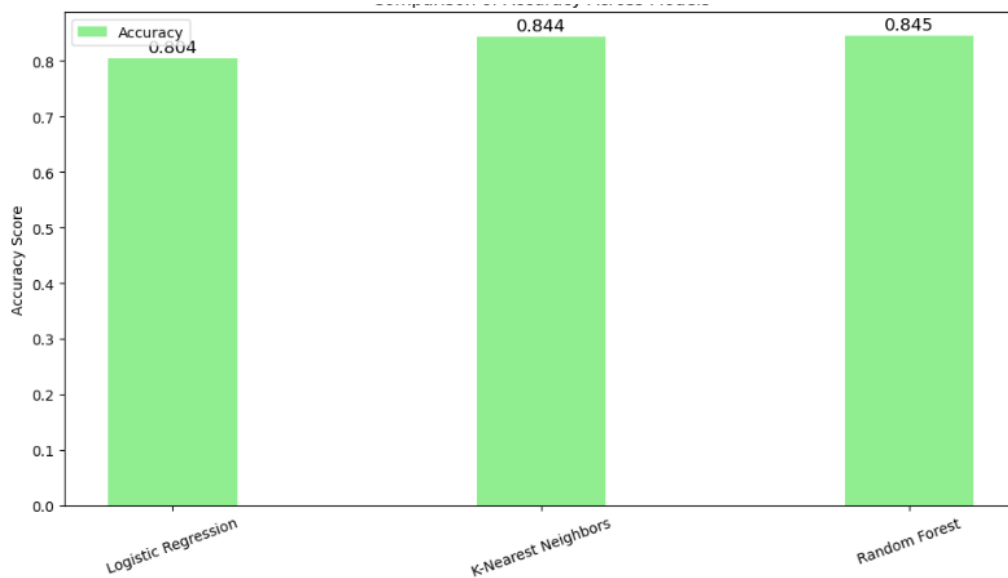


Figure 19 Comparison of Accuracy across models after 10-fold cross validation

Figure 16, the accuracy comparison across three models after using 10-fold cross validation presented that K-Nearest Neighbors (KNN) and Random Forest had the best performance. Logistic Regression achieves an accuracy of approximately 0.804, indicating solid baseline performance but falling behind the other two models. KNN improves on this result with an accuracy of about 0.844, demonstrating that it captures more complex relationships in the data. Random Forest slightly surpasses KNN with the highest accuracy of 0.845, making it the best-performing model overall. The difference between KNN and Random Forest is small, but Random Forest's slight edge suggests that its ability to combine multiple decision trees provides more stable and reliable predictions. Overall, the results indicate that ensemble and distance-based methods outperform linear classification for this dataset, with Random Forest offering the most accurate performance across folds.

PART VII: Final Presentation Videos: YouTube Links

<https://youtu.be/bv-v-UqmB-o>